# Do Now · retrieval

## Software testing: finding and correcting faults

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

Define a cache miss.

### Q2

When is A AND B true?

### Q3

Give a boundary-invalid input just below 1.

### Q4

Does 61 satisfy `seconds>=1 || seconds<=60`?

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Read the stated rule — Because the permitted ages are inclusive, both age 12 and age 18 must be accepted.

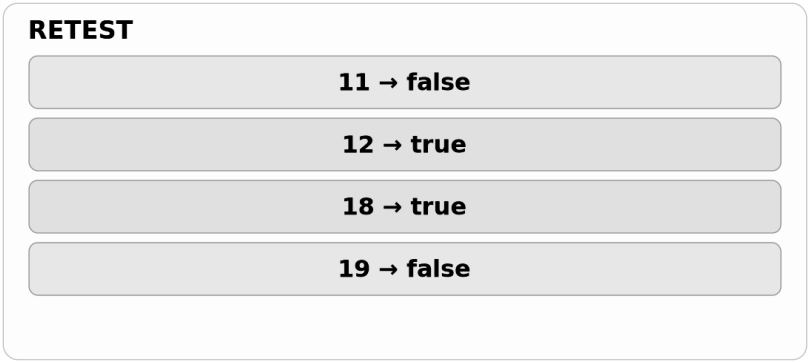
## Core idea 2

Locate the Boolean error — The strict comparison operators > and < exclude both valid boundaries.

## Core idea 3

Repair and test — Change to >= and <=; retest the boundaries plus values just outside the range.

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.

# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

logic error — Runs but produces the wrong result.

inclusive boundary — Endpoint allowed by a range.

validation — Check input meets constraints.

debugging — Find and correct a fault.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

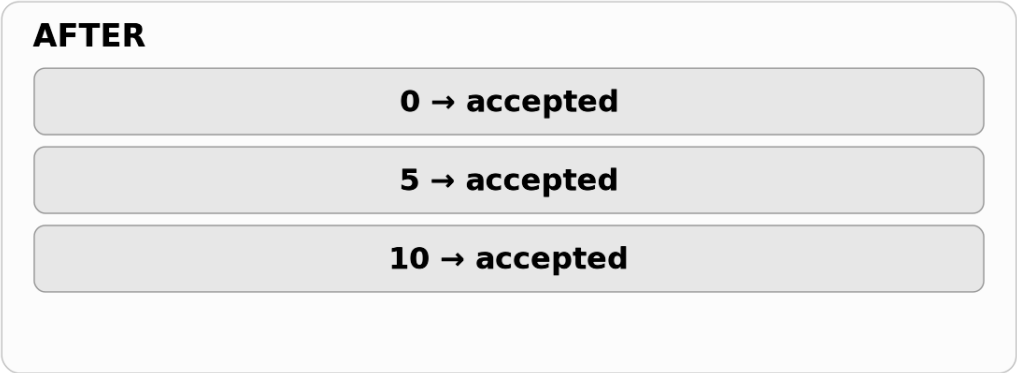
## Problem

A score validator intended for 0–10 uses score>0 && score<10. Fix it.

## Follow the worked method

1. Predict first — The valid endpoints were incorrectly rejected.
2. Model and check — Change the operators to >= and <=, and test −1 and 11 as invalid values.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.

# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Predict first
OR accepts 61 because only one side needs to be true.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Correct x>5 && x<15 for values 5–15 inclusive, and state whether 5 and 15 are valid.

## Question 2

Explain why `age>=12 || age<=18` accepts 100, correct it, and give outputs for 11,12,18,19.

Check that you have used the technical terms and explained your decisions.

# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

Outline a debugging and regression-testing process after correcting a validator.

Check that you have used the technical terms and explained your decisions.

# Exit Ticket · reflect and retrieve

## Exit Ticket

For a bounded range, OR usually replaces AND without changing the outcome.

## Use key vocabulary from memory

1. Explain logic error in your own words.

2. Explain inclusive boundary in your own words.

3. Explain validation in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.